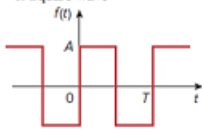
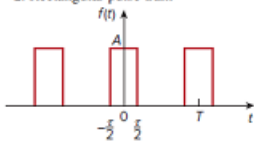
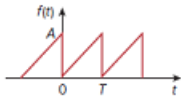
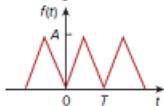
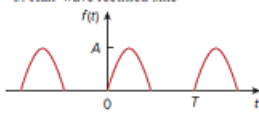
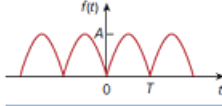


Problem

Design a low-pass RC filter with a resistance $R = 2\text{ k}\Omega$. The input to the filter is a periodic rectangular pulse train (see Table 17.3) with $A = 1\text{ V}$, $T = 10\text{ ms}$ and $\tau = 1\text{ ms}$. Select C such that the dc component of the output is 50 times greater than the fundamental component of the output.

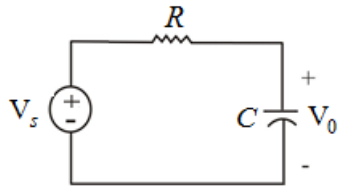
Table 17.3: The Fourier series of common functions.

Function	Fourier series
1. Square wave 	$f(t) = \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\omega_0 t$
2. Rectangular pulse train 	$f(t) = \frac{A\tau}{T} + \frac{2A}{T} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi\tau}{T} \cos n\omega_0 t$
3. Sawtooth wave 	$f(t) = \frac{A}{2} - \frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\omega_0 t}{n}$
4. Triangular wave 	$f(t) = \frac{A}{2} - \frac{4A}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\omega_0 t$
5. Half-wave rectified sine 	$f(t) = \frac{A}{\pi} + \frac{A}{2} \sin \omega_0 t - \frac{2A}{\pi} \sum_{n=2}^{\infty} \frac{1}{4n^2-1} \cos 2n\omega_0 t$
6. Full-wave rectified sine 	$f(t) = \frac{2A}{\pi} - \frac{4A}{\pi} \sum_{n=2}^{\infty} \frac{1}{4n^2-1} \cos n\omega_0 t$

Step-by-step solution

Step 1 of 4

Draw the low pass RC circuit.



Step 2 of 4

The input to the filter is,

$$V_s = \frac{A\tau}{T} + \frac{2A}{T} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi\tau}{T} \cos n\omega_o t \dots\dots (1)$$

From the circuit, the output voltage is,

$$V_o = \frac{\frac{1}{j\omega_n C}}{R + \frac{1}{j\omega_n C}} V_s$$

$$= \left(\frac{1}{1 + j\omega_n RC} \right) V_s$$

Here,

$$\omega_n = n\omega_o$$

$$= \frac{2n\pi}{T}$$

Step 3 of 4

For $n = 0$,

$$V_o = \left(\frac{1}{1 + j(0)RC} \right) V_s$$

$$= V_s$$

From equation (1),

$$V_s = \frac{A\tau}{T}$$

Therefore,

$$V_o = \frac{A\tau}{T} \dots\dots (2)$$

For the n^{th} harmonic,

$$V_o = \frac{1}{\sqrt{1 + \omega_n^2 R^2 C^2}} \angle \tan^{-1} \omega_n RC \frac{2A}{nT} \sin \frac{n\pi\tau}{T} \angle -90^\circ$$

Calculate the magnitude of output when $n = 1$.

$$|V_o| = \frac{2A}{T} \sin \frac{\pi\tau}{T} \left(\frac{1}{\sqrt{1 + \frac{4\pi^2}{T^2} R^2 C^2}} \right) \dots\dots (3)$$

Step 4 of 4

The dc component of the output is 50 times greater than the fundamental component of the output.

That is,

$$\frac{A(1 \times 10^{-3})}{T} = 50 \times \frac{2A}{T} \sin \frac{\pi(1 \times 10^{-3})}{10 \times 10^{-3}} \frac{1}{\sqrt{1 + \frac{4\pi^2}{T} R^2 C^2}}$$

$$1 \times 10^{-3} = 100 \sin \frac{\pi}{10} \frac{1}{\sqrt{1 + \frac{4\pi^2}{T} R^2 C^2}}$$

$$\sqrt{1 + \frac{4\pi^2}{T} R^2 C^2} = 3.09 \times 10^4$$

$$1 + \frac{4\pi^2}{T} R^2 C^2 = 954.92 \times 10^6$$

$$\frac{4\pi^2}{(10 \times 10^{-3})} (2000)^2 C^2 = 954.92 \times 10^6$$

$$1.579 \times 10^{10} C^2 = 954.92 \times 10^6$$

$$C^2 = 60.47 \times 10^{-3}$$

$$C = 245.9 \text{ mF}$$

Therefore, the value of the capacitor, C is 245.9 mF.